

● MEDIUM

● ADVANCED MATH

● ANSWER KEY

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# Advanced Math

# 04

10 answers · Rationale-rich · Teach from doc

**Q1****11875**

The correct answer is 11,875. It's given that the exponential function  $g$  is defined by  $gx = 19 \cdot a^x$ , where  $a$  is a positive constant, and  $g3 = 2,375$ . It follows that when  $x = 3$ ,  $gx = 2,375$ .

Substituting 3 for  $x$  and 2,375 for  $gx$  in the given equation yields  $2,375 = 19 \cdot a^3$ . Dividing each side of this equation by 19 yields  $125 = a^3$ . Taking the cube root of both sides of this equation gives  $a = 5$ . Substituting 4 for  $x$  and 5 for  $a$  in the equation  $gx = 19 \cdot a^x$  yields  $g4 = 19 \cdot 5^4$ , or  $g4 = 11,875$ . Therefore, the value of  $g4$  is 11,875.

**Q2****D**

D.  $x = by - 72y$

Choice D is correct. Multiplying both sides of the given equation by  $y$  yields  $y(b - 72) = x$ .

Distributing on the left-hand side of this equation yields  $by - 72y = x$ , or  $x = by - 72y$ . Therefore, the equation  $x = by - 72y$  correctly expresses  $x$  in terms of  $b$  and  $y$ .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

**Q3****B**

B.  $2x^2 - xy - 3y^2$

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Choice B is correct. Expanding all terms yields  $(x - 11y)(2x - 3y) - 12y(-2x + 3y)$ , which is equivalent to  $2x^2 - 22xy - 3xy + 33y^2 + 24xy - 36y^2$ . Combining like terms gives  $2x^2 - xy - 3y^2$ .

Choice A is incorrect and may be the result of using the sums of the coefficients of the existing  $x$  and  $y$  terms as the coefficients of the  $x$  and  $y$  terms in the new expressions. Choice C is incorrect and may be the result of incorrectly combining like terms. Choice D is incorrect and may be the result of using the incorrect sign in front of the  $12y$  term.

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**Q4****D**

- D. The estimated number of catalogs the company sent to its customers at the end of 1992 was 9,000.
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Choice D is correct. The  $y$ -intercept of the graph is the point at which the graph crosses the  $y$ -axis, or the point for which the value of  $x$  is 0. Therefore, the  $y$ -intercept of the given graph is the point  $(0, 9)$ . It's given that  $x$  represents the number of years since the end of 1992. Therefore,  $x = 0$  represents 0 years since the end of 1992, which is the same as the end of 1992. It's also given that  $y$  represents the estimated number of catalogs, in thousands, that the company sent to its customers at the end of the year. Therefore,  $y = 9$  represents 9,000 catalogs. It follows that the  $y$ -intercept  $(0, 9)$  means that the estimated number of catalogs the company sent to its customers at the end of 1992 was 9,000.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

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**Q5****A**

A.  $T = \frac{mv^2}{L}$

Choice A is correct. To write the formula as T in terms of m, v, and L means to isolate T on one side of the equation. First, multiply both sides of the equation by m, which gives  $mv^2 = \frac{mLT}{m}$ , which simplifies to  $mv^2 = LT$ . Next, divide both sides of the equation by L, which gives  $\frac{mv^2}{L} = \frac{LT}{L}$ , which simplifies to  $T = \frac{mv^2}{L}$ .

Choices B, C, and D are incorrect and may be the result of incorrectly applying operations to each side of the equation.

**Q6****C**

C.  $x+10$

Choice C is correct. Distributing the negative sign to the terms in the second parentheses yields  $(2x+3)-x+7$ . This expression can be rewritten as  $2x-x+3+7$ . Combining like terms results in  $x+10$ .

Choice A is incorrect and may result from not distributing the negative sign to the 7. Choice B is incorrect and may result from adding  $(x-7)$  to  $2x+3$  instead of subtracting  $(x-7)$ . Choice D is incorrect and may result from adding the product of  $2x$  and x to the product of 3 and 7.

## Q7

D

D.  $fx = x^2 + 5x + 14$

Choice D is correct. The equation of a quadratic function can be written in the form  $fx = ax - h^2 + k$ , where  $a$ ,  $h$ , and  $k$  are constants. It's given in the table that when  $x = -1$ , the corresponding value of  $fx$  is 10. Substituting  $-1$  for  $x$  and 10 for  $fx$  in the equation  $fx = ax - h^2 + k$  gives  $10 = a(-1) - h^2 + k$ , which is equivalent to  $10 = -a - h^2 + k$ , or  $10 = -a + 2ah + ah^2 + k$ . It's given in the table that when  $x = 0$ , the corresponding value of  $fx$  is 14. Substituting 0 for  $x$  and 14 for  $fx$  in the equation  $fx = ax - h^2 + k$  gives  $14 = a(0) - h^2 + k$ , or  $14 = -ah^2 + k$ . It's given in the table that when  $x = 1$ , the corresponding value of  $fx$  is 20. Substituting 1 for  $x$  and 20 for  $fx$  in the equation  $fx = ax - h^2 + k$  gives  $20 = a(1) - h^2 + k$ , which is equivalent to  $20 = a - h^2 + k$ , or  $20 = a - 2ah + ah^2 + k$ . Adding  $20 = a - 2ah + ah^2 + k$  to the equation  $10 = -a + 2ah + ah^2 + k$  gives  $30 = 2a + 2ah^2 + 2k$ . Dividing both sides of this equation by 2 gives  $15 = a + ah^2 + k$ . Since  $14 = -ah^2 + k$ , substituting 14 for  $-ah^2 + k$  into the equation  $15 = a + ah^2 + k$  gives  $15 = a + 14$ . Subtracting 14 from both sides of this equation gives  $a = 1$ . Substituting 1 for  $a$  in the equations  $14 = -ah^2 + k$  and  $20 = a - h^2 + k$  gives  $14 = -h^2 + k$  and  $20 = 1 - h^2 + k$ , respectively. Since  $14 = -h^2 + k$ , substituting 14 for  $-h^2 + k$  in the equation  $20 = 1 - h^2 + k$  gives  $20 = 1 - 14 + k$ , or  $20 = -13 + k$ . Adding 13 to both sides of this equation gives  $33 = k$ . Substituting 1 for  $a$  and 33 for  $k$  in the equation  $fx = ax - h^2 + k$  gives  $fx = x - h^2 + 33$ , which is equivalent to  $fx = x^2 + 5x + 14$ . Therefore,  $fx = x^2 + 5x + 14$  defines  $f$ .

Choice A is incorrect. If  $fx = 3x^2 + 3x + 14$ , then when  $x = -1$ , the corresponding value of  $fx$  is 14, not 10.

Choice B is incorrect. If  $fx = 5x^2 + x + 14$ , then when  $x = -1$ , the corresponding value of  $fx$  is 18, not 10.

Choice C is incorrect. If  $fx = 9x^2 - x + 14$ , then when  $x = -1$ , the corresponding value of  $fx$  is 24, not 10, and when  $x = 1$ , the corresponding value of  $fx$  is 22, not 20.

**Q8****6**

The correct answer is 6. It's given that  $y = 4x$  and  $y = x^2 - 12$ . Since  $y = 4x$ , substituting  $4x$  for  $y$  in the second equation of the given system yields  $4x = x^2 - 12$ . Subtracting  $4x$  from both sides of this equation yields  $0 = x^2 - 4x - 12$ . This equation can be rewritten as  $0 = x - 6x + 12$ . By the zero product property,  $x - 6 = 0$  or  $x + 2 = 0$ . Adding 6 to both sides of the equation  $x - 6 = 0$  yields  $x = 6$ . Subtracting 2 from both sides of the equation  $x + 2 = 0$  yields  $x = -2$ . Therefore, solutions to the given system of equations occur when  $x = 6$  and when  $x = -2$ . It's given that a solution to the given system of equations is  $x, y$ , where  $x > 0$ . Since 6 is greater than 0, it follows that the value of  $x$  is 6.

**Q9****B**

B.  $(x + 3)^2 - 5$

Choice B is correct. The given quadratic expression is in standard form, and each answer choice is in vertex form. Completing the square converts the expression from standard form to vertex form. The first step is to rewrite the expression as follows:  $x^2 + 6x + 4 = x^2 + 6x + 9 + 4 - 9$ .

The first three terms of the revised expression can be rewritten as a perfect square as follows:  $x^2 + 6x + 9 + 4 - 9 = (x + 3)^2 + 4 - 9$ . Combining the constant terms gives  $(x + 3)^2 - 5$ .

Choice A is incorrect. Squaring the binomial and simplifying the expression in choice A gives  $x^2 + 6x + 9 + 5$ . Combining like terms gives  $x^2 + 6x + 14$ , not  $x^2 + 6x + 4$ . Choice C is incorrect. Squaring the binomial and simplifying the expression in choice C gives  $x^2 - 6x + 9 + 5$ . Combining like terms gives  $x^2 - 6x + 14$ , not  $x^2 + 6x + 4$ . Choice D is incorrect. Squaring the binomial and simplifying the expression in choice D gives  $x^2 - 6x + 9 - 5$ . Combining like terms gives  $x^2 - 6x + 4$ , not  $x^2 + 6x + 4$ .

## Q10

D

D.  $w + 9$

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Choice D is correct. It's given that the expression  $ww + 9$  represents the area, in square centimeters, of a rectangular cutting board, where  $w$  is the width, in centimeters, of the cutting board. The area of a rectangle can be calculated by multiplying its length by its width. It follows that the length, in centimeters, of the cutting board is represented by the expression  $w + 9$ .

Choice A is incorrect. This expression represents the area, in square centimeters, of the cutting board, not its length, in centimeters.

Choice B is incorrect. This expression represents the width, in centimeters, of the cutting board, not its length.

Choice C is incorrect. This is the difference between the length, in centimeters, and the width, in centimeters, of the cutting board, not its length, in centimeters.

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### Notes

Accept equivalent forms (e.g.  $0.25 \equiv 1/4$ ). For grid-in items, decimal and fraction answers are both valid.

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